

Problem 11.27

Fund X starts with 1,000 and accumulates with a force of interest

$$\delta_t = \frac{1}{15-t}, \quad 0 \leq t < 15.$$

Fund Y starts with 1,000 and accumulates with an interest rate of 8% per annum compounded semiannually for the first three years and an effective interest rate of i per annum thereafter. Fund X equals Fund Y at the end of four years. Calculate i .

Solution

The value of Fund X at the end of four years is

$$\begin{aligned} 1,000 \exp \left(\int_0^4 \delta_t dt \right) &= 1,000 \exp \left(\int_0^4 \frac{1}{15-t} dt \right) \\ &= 1,000 \exp (\ln(15) - \ln(11)) \\ &= 1,000 \left(\frac{15}{11} \right) \\ &\approx 1,363.636364. \end{aligned}$$

For Fund Y , the first three years have six semiannual periods at a nominal rate of 8%. The fourth year earns the effective rate i . Thus, its value at the end of four years is

$$1,000 \left(1 + \frac{0.08}{2} \right)^6 (1 + i).$$

Equating the fund values and solving for i ,

$$\begin{aligned} 1,000 \left(1 + \frac{0.08}{2} \right)^6 (1 + i) &= 1,000 \left(\frac{15}{11} \right), \\ i &= \frac{15}{11 \left(1 + \frac{0.08}{2} \right)^6} - 1 \\ &\approx 0.077701626. \end{aligned}$$

Therefore, the annual effective interest rate is

$$\boxed{i \approx 0.077701626 = 7.7702\%}.$$